SOFTWARE REQUIREMENTS DOCUMENT

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Project: Student Records Manager | Week 1: 11-18 October 2026
Version: 1.0 | Status: Week 1 requirements baseline

1. Purpose and Scope

This document defines what Student Records Manager must do (functional requirements) and how it should behave (non-functional requirements). It supports a six-week web application for two staff roles: Administrator and Teacher. The seven features match the accompanying project proposal.

2. Functional Requirements

ID	Requirement	Expected behavior
FR-01	Authentication	Authorized administrator and teacher users shall be able to log in and log out.
FR-02	Authorization	The system shall enforce access based on role: administrators manage student records and user access; teachers view records and manage marks within permissions.
FR-03	Student CRUD	Authorized users shall create, view, update and delete student records containing name, roll number, class and contact information, subject to role permissions.
FR-04	Search	Users shall search for student records using a name or roll number.
FR-05	Marks	Authorized staff shall enter, view and update subject-wise student marks.
FR-06	Results	The system shall calculate total obtained marks, percentage and grade using a documented grading rule.
FR-07	Dashboard	The system shall display total student count and basic academic performance summaries.
FR-08	Validation	The system shall reject missing required values, invalid marks and duplicate roll numbers according to the uniqueness rule selected during database design.

3. Non-Functional Requirements

ID	Quality	Requirement
NFR-01	Security	Passwords shall be hashed; protected server endpoints shall verify authentication and authorization.
NFR-02	Usability	Forms and navigation shall be understandable, with clear success/error feedback.
NFR-03	Performance	Common user actions should complete within about 3 seconds in a normal development/demo test environment; measurement conditions shall be recorded.
NFR-04	Integrity	Input shall be validated before saving, and student/marks relationships shall remain consistent.
NFR-05	Maintainability	Code shall use distinct frontend, backend, model and test areas, with meaningful names and basic documentation.
NFR-06	Reliability	Expected invalid inputs and application errors shall return helpful responses rather than crash the application.
NFR-07	Compatibility	The web interface should work in current major desktop browsers.

4. User Stories

US-01: As an administrator, I want to add student records so that information is stored digitally.

US-02: As an administrator, I want to edit or remove student records so that incorrect or outdated details can be corrected.

US-03: As a teacher, I want to search by name or roll number so that I can locate a student quickly.

US-04: As a teacher, I want to record subject marks so that student academic progress is documented.

US-05: As a teacher, I want automatic totals, percentages and grades so that manual calculation errors are reduced.

US-06: As an administrator, I want role-based access so that only authorized staff can perform restricted actions.

US-07: As an administrator, I want a dashboard summary so that I can review student and academic statistics.

US-08: As an authorized staff member, I want to log in and out securely so that access to records is controlled.

5. Design Decisions to Finalize in Later Weeks

The exact grade thresholds, roll-number uniqueness scope (institution-wide or class-wise), administrator provisioning method, and teacher permission boundaries will be defined during Week 2 database/application design and Week 4 authentication implementation. These decisions must be documented and kept consistent with this baseline.

6. Exclusions

Online payments, attendance, student/parent portals, automated SMS or email notifications and advanced predictive analytics are excluded from this six-week scope.

7. Roadmap Alignment

Week 1 defines requirements; Week 2 converts entities and relationships to database design and an ER diagram; Week 3 implements core CRUD and calculation logic; Week 4 adds authentication and appropriate advanced features; Week 5 documents testing and fixes; Week 6 delivers the final application, documentation and presentation.

Document status: Requirements baseline prepared for Week 1 submission. Functional correctness will be validated during implementation and testing.